

Course: B. Tech. Branch : Computer Engineering/Computer Science and Engineering

Subject Code & Name: Database Systems (BTCOC501)

Semester :V

Max Marks: 60

Date:01/07/2024

Duration: 3 Hr.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

(Level/CO) Marks

Q. 1 Solve Any Two of the following.		12
A) Draw and explain the detailed system architecture of DBMS.	Understand	6
B) Explain in detail about various key constraints used in database system.	Understand	6
C) Discuss the main characteristics of the database approach and specify how it differs from traditional file system?	Analysis	6
Q.2 Solve Any Two of the following.		12
A) Write a short notes on i) Foreign Key ii) Relation state iii) Database schema.	Understand	6
B) account(account_number, branch_name, balance) branch (branch_name, branch_city, assets) customer (customer_name ,customer_street, customer_city) loan (loan_number, branch_name, amount) depositor((customer_name, account_number) borrower(customer_name, loan_number) Write the following queries in SQL: 1)For all customers who have a loan from the bank, find their names, loan numbers, and loan amount. 2) Find the customer names, loan numbers, and loan amounts, for all loans at the Panvel branch. 3)Find the names of all branches that have assets greater than those of at least one branch located in Mumbai. 4)Find the average account balance of those branches where the account balance is greater than Rs. 1500. 5) Find the maximum across all branches of the total balance at each branch.	Apply	6
C) Write the SQL syntax for the following with example:	Understand	6

1)SELECT 2) ALTER 3)UPDATE

Q. 3	Solve Any Two of the following.		12
A)	Write an SQL query for the following:	Apply	6
	a)To create a table of Hospital database with minimum 4 fields		
	b) To insert two records		
	c) To add new field		
	d)To display all records		
B)	What are JOINS? Explain INNER JOIN and OUTER JOIN.	Understand	6
C)	Explain different types of trigger.	Understand	6
Q.4	Solve Any Two of the following.		12
A)	State BCNF. How does it differ from 3NF?	Analysis	6
B)	Explain about dynamic multilevel indexing using B+ trees.	Understand	6
C)	Define Multi-valued dependency. Explain the Fourth normal form with an example.	Understand	6
Q. 5	Solve Any Two of the following.		12
A)	Explain in detail about timestamp based concurrency control techniques.	Understand	6
B)	Explain ACID properties of a transaction.	Understand	6
C)	Explain how Concurrency control can be achieved with locking methods?	Understand	6

*** End ***